COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structure and Analysis	**SECTION:**	CPE21S2
DATE PERFORMED:	Jul 17, 2026	**DATE SUBMITTED:**	Jul 17, 2026
NAME:	Antonio, Christine Mae P.	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

HOA NO. 2.1 Arrays, Pointers and Dynamic Memory Allocation

1. PROCEDURE OUTPUT

What procedures did you perform?

Screenshot

```cpp
#include <iostream>
#include <string.h>
using namespace std;

class Student {
    private:
        string studentName;
        int studentAge;

    public:
        //constructor
        Student(string newName = "John Doe", int newAge = 18) {
            studentName = move(newName);
            studentAge = newAge;
            cout << "Constructor called." << endl;
        }

        //Destructor
        ~Student() {
            cout << "Destructor Called." << endl;
        }

        //Copy Constructor
        Student(const Student &copyStudent) {
            cout << "Copy Constructor Called." << endl;
            studentName = copyStudent.studentName;
            studentAge = copyStudent.studentAge;
        }

        //Display Attributes
        void printDetails() {
            cout << this->studentName << " " << this->studentAge << endl;
        }
};

int main() {

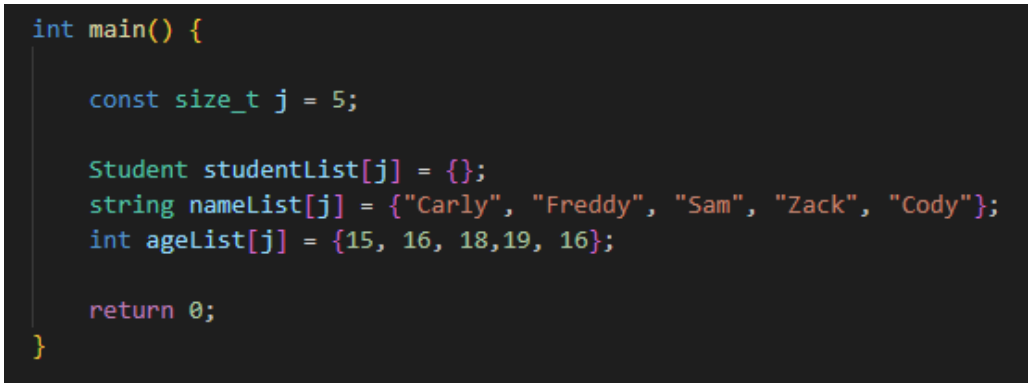
    Student student1("Roman", 28);
    Student student2(student1);
    Student student3;
    student3 = student2;

    return 0;
}
```

```
Constructor called.
Copy Constructor Called.
Constructor called.
Destructor Called.
Destructor Called.
Destructor Called.
```

Observation	The Student class was created with private and public access modifiers. The public access modifier contains the constructor, destructor, copy constructor, copy assignment operator, and printDetails() function. The constructor have arguments newName (string) and newAge (int), with initial values of “John Doe” and 18, respectively. Inside the constructor sets the values of newName and newAge to studentName and studentAge, respectively. The destructor will simply print “Destructor called.” when the program will end.

Table 2-1. Initial Driver Program

Screenshot	
Observation	

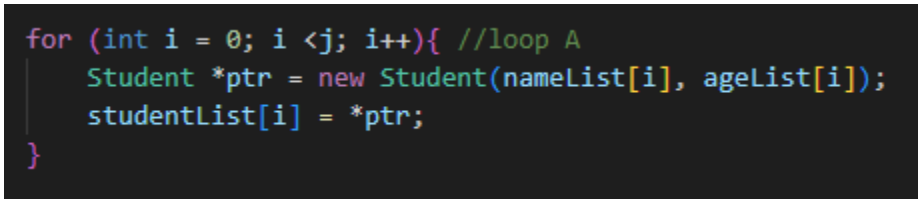
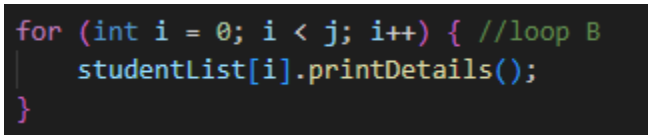
```
int main() {

    const size_t j = 5;

    Student studentList[j] = {};
    string nameList[j] = {"Carly", "Freddy", "Sam", "Zack", "Cody"};
    int ageList[j] = {15, 16, 18,19, 16};

    return 0;
}
```

Table 2-2. Modified Driver Program with Student Lists

Loop A	
Observation	
Loop B	
Observation	

```
for (int i = 0; i <j; i++){ //loop A
    Student *ptr = new Student(nameList[i], ageList[i]);
    studentList[i] = *ptr;
}
```

```
for (int i = 0; i < j; i++) { //loop B
    studentList[i].printDetails();
}
```

Output	Constructor called. Constructor called. Constructor called. Constructor called. Constructor called. Constructor called. Constructor called. Constructor called. Constructor called. Constructor called. Carly 15 Freddy 16 Sam 18 Zack 19 Cody 16 Destructor Called. Destructor Called. Destructor Called. Destructor Called. Destructor Called.
Observation	

Table 2-3. Final Driver Program

Modification	1.
	`//Copy Assignment Operator`
	`Student& operator=(const Student& copy) {`
	`  cout << "Copy Assignment Operator Called." << endl;`
	`  studentName = copy.studentName;`
	`  studentAge = copy.studentAge;`
	`  return *this;`
	`}`
	`int main() {`
	`  Student student1("Roman", 28);`
	`  Student student2(student1);`
	`  Student student3;`
	`  student3 = student2;`
	`  student1.printDetails();`
	`  student2.printDetails();`
	`  student3.printDetails();`
	`  return 0;`
	`}`
	Constructor called.
	Copy Constructor Called.
	Constructor called.
	Copy Assignment Operator Called.
	Roman 28
	Roman 28
	Roman 28
	Destructor Called.
	Destructor Called.
	Destructor Called.

	2.
Observation	

Table 2-4. Modifications/Corrections Necessary

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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